

Code-Layout, Readability and Reusability

Date	28 September 2026
Team ID	SWTID-2026-7718
Project Name	Ai Study Buddy
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Code uses consistent indentation and spacing.
2	Proper File Structure	Files and folders are logically organized	Yes	Files and folders are logically organized.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables have clear and meaningful names.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named according to their purpose.
5	Code Comments	Inline and block comments explain logic	Yes	Comments are added where necessary to explain the code.
6	Modular Design	Code is split into reusable functions/modules	Yes	The project is divided into reusable modules/functions.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unnecessary and duplicate code is avoided.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Basic errors and exceptions are handled properly.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	User Interface Module	HTML, CSS, JavaScript	Login, study dashboard and interaction pages	High

2	Question Generation Module	Python	Generating practice questions and quizzes	High
3	Study Material Processing Module	Python	Processing uploaded study materials	High
4	AI Response Module	Python / AI API	Providing explanations and answers	High
5	User Data Module	Python / Database	Storing and retrieving user study information	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Code is well-organized with a clear structure.
Readability	5	Code is simple, clear, and easy to understand.
Reusability	4	Modules and functions can be reused in different parts of the project.
Documentation / Comments	4	Comments and documentation explain the important parts of the code
Overall Score	5	The project has good code quality, structure, readability, and maintainability.